

Time: 2 hours.

In the following $A, B, \dots$ are propositional variables, $a, b, \dots$ constant symbols, $f, g, \dots$ function symbols, $X, Y, \dots$ variables, $p, q, \dots$ predicate symbols and $F, G, \phi, \psi, \dots$ formulas (unless differently specified).

1. Consider the language of propositional logic. Use natural deduction to prove that the following holds, or find a counter-example to show that it does not hold

- $(A \rightarrow (B \rightarrow C)) \rightarrow ((A \wedge B) \rightarrow C)$
- $(F \rightarrow G) \rightarrow (\neg G \rightarrow F)$

2. (4 points) Prove that that $\phi \models \psi$ (ψ is a logical consequence of ϕ) or that $\phi \not\models \psi$ for the following formulas:

- $\phi : (A \rightarrow B) \wedge (A \rightarrow C)$ and $\psi : \neg B \vee \neg C \vee A$
- $\phi : A \vee B \vee \neg C$ and $\psi : (A \wedge B) \rightarrow C$

3. A robbery took place in Wherevercity. Investigators found that if the robbery occurred at midnight then surely the robber(s) fled the city. If the robbery didn't happen at midnight then it involved two robbers. The sheriff of Wherevercity claims that the robber is out of the city or there were two of them. Formalize the situation with propositional logic and establish whether the three statements are contradictory. Provide a short motivation for your answer.

4. Let us consider the language $\mathcal{L} = \{Y, C, U, W, K\}$ where Y, C, U, W are unary relational symbols ($Y(x)$ stands for "x is under 30 years old", $G(x)$ stands for "x is a girl", $C(x)$ stands for "x is enrolled in college", $U(x)$ stands for "x can use a computer", $W(x)$ stands for "x will find work"), while K is a binary relational symbol ($K(x, y)$ stands for "x knows y"). Formalize the following sentences:

- (a) All people under 25 years old who is enrolled in college know how to use a computer or know someone who can.
- (b) Among those who don't know how to use a computer there are some who will find work.
- (c) Everyone knows someone under 25 years old who will find work.

5. Professor X has published all the exams' dates for the whole year at once. Even though this looks like a student-friendly measure, his ultimate goal is to write the lowest number of exercises. In fact, Professor X's exam contains two exercises, A and B, which he wants to recycle between two dates. In order for him to do so, no student should attend both dates but that is unfortunately what Chris, Diana and Ethan have done. By carefully examining the student lists of the 5 available exams, X notices that:

- Chris is in the list for date 1, 3, and 5.
- Diana is in the list for date 1 and 2.
- Ethan is in the list for date 2 and 4.

Additionally, to avoid rumors of his exam always looking the same, Professor X does not want any exercise to appear in the same position twice (e.g. exercise 1 cannot appear twice in position A). Will X manage to prepare all the exams using only 6 exercises?

If that is the case, how can the three students be sure to make sure that, if Professor X only uses 6 exercises, one of them will for sure find the same exercise twice? To this end, note that a student can attend a maximum of 3 dates. Model the problem using CLP and show a possible solution.

6. Define a Prolog predicate that creates the list of integer numbers with first and last element given.

For example:

```
?- create_list(-1, 7, X).
```

```
X = [-1, 0, 1, 2, 3, 4, 5, 6, 7].
```

Note that you can use the predicates seen in the course and the predicate `last/2`, that returns the last element of a given list:

```
?- last([1,2,3], X).
```

```
X = 3.
```

ES. 1

- 1) $(A \rightarrow (B \rightarrow C)) \rightarrow ((A \wedge B) \rightarrow C) \stackrel{?}{=} \perp$ so it's always true
- a) $(A \rightarrow (B \rightarrow C)) \equiv T$ so $T \rightarrow (T \rightarrow \perp) \equiv T \rightarrow \perp \equiv \perp \neq T$
- b) $((A \wedge B) \rightarrow C) \equiv \perp$ so $C \equiv \perp, A \equiv T, B \equiv T$

- 2) $(F \rightarrow G) \rightarrow (\neg G \rightarrow F) \stackrel{?}{=} \perp$ so it can be false when F and G are false

- 1) $F \rightarrow G \equiv T$ so $\perp \rightarrow \perp \equiv T$
- 2) $\neg G \rightarrow F \equiv \perp$ so $F \equiv \perp$
- $\neg G \equiv T \quad G \equiv \perp$

ES. 2

- 1) $[(A \rightarrow B) \wedge (A \rightarrow C)] \rightarrow [\neg B \vee \neg C \vee A] \stackrel{?}{=} \perp \Rightarrow$ so it can be false (not logical consequence) when

- a) $(A \rightarrow B) \wedge (A \rightarrow C) \equiv T$ so $(\perp \rightarrow T) \wedge (\perp \rightarrow T) \equiv T$ B and C are true and A is false
- b) $[\neg B \vee \neg C \vee A] \equiv \perp$ so $B \equiv T, C \equiv T, A \equiv \perp$

- 2) $[A \vee B \vee \neg C] \rightarrow [(A \wedge B) \rightarrow C] \stackrel{?}{=} \perp$ so it can be false (not logical consequence) when C is false

- a) $[A \vee B \vee \neg C] \equiv T$ so $T \vee T \vee T \equiv T$ an the others are true
- b) $[(A \wedge B) \rightarrow C] \equiv \perp$ so $C \equiv \perp, A \equiv T, B \equiv T$

ES. 3

A = robbery at midnight

B = robbery's fled the city

C = there were 2 robbers

1) $A \rightarrow B$

2) $\neg A \rightarrow C$

3) $B \vee C$

Is contradictory? No if it is satisfiable so we look for

① $\wedge$ ② $\wedge$ ③ $\equiv T$

$(A \rightarrow B) \wedge (\neg A \rightarrow C) \wedge (B \vee C) \equiv T$

$(\neg A \vee B) \wedge (A \vee C) \wedge (B \vee C) \equiv T$

$\swarrow \neg A \vee B \equiv T$
 $\rightarrow A \vee C \equiv T$
 $\searrow B \vee C \equiv T$

$\nearrow$ it's not contradictory
 it's true when A, B, C are true

ES. 4

$Y(x) = x$ is under 30 $U(x) = x$ can use computer
 $G(x) = x$ is girl $W(x) = x$ will find work
 $C(x) = x$ is in college $K(x,y) = x$ knows y

a) $\forall x: [Y(x) \wedge C(x)] \longrightarrow U(x) \vee [\exists y: K(x,y) \wedge U(y)]$

b) $\exists x: \neg U(x) \wedge W(x)$

c) $\forall x: \exists y: K(x,y) \wedge Y(y) \wedge W(y)$

ES. 5

exam (VARIABLES) :-

length(VARIABLES, 10),

VARIABLES = [A1, B1, A2, B2, A3, B3, A4, B4, A5, B5],

VARIABLES ins 1..6,

all_different([A1, B1, A2, B2]),

all_different([A4, B4, A2, B2]),

all_different([A1, B1, A3, B3, A5, B5]),

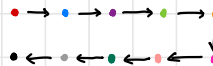
all_different([A1, A2, A3, A4, A5]),

all_different([B1, B2, B3, B4, B5])

From prolog: VARIABLES = $\frac{1}{1}, \frac{2}{2}, \frac{3}{3}, \frac{4}{4}, \frac{3}{5}, \frac{2}{4}, \frac{1}{5}, \frac{5}{6}$

Showing a possible solution without prolog:

$B1 :: [1, 2, 3, 4, 5, 6]$ $A1 :: [2, 3, 4, 5, 6]$
 $B2 :: [1, 2, 3, 4, 5, 6]$ $A2 :: [1, 2, 3, 4, 5, 6]$
 $B3 :: [1, 2, 3, 4, 5, 6]$ $A3 :: [1, 2, 3, 4, 5, 6]$
 $B4 :: [1, 2, 3, 4, 5, 6]$ $A4 :: [1, 2, 3, 4, 5, 6]$
 $B5 :: [1, 2, 3, 4, 5, 6]$ $A5 :: [1, 2, 3, 4, 5, 6]$



SOL = [1, 2, 3, 4, 4, 3, 2, 1, 5, 6]

ES. 6

$\text{create_list}(E, E, [E]) :- !.$

$\text{create_list}(N, N2, [N|Res]) :-$

$N < N2,$

$N1 \text{ is } N+1,$

$\text{create_list}(N1, N2, Res).$